

BIG DATA ANALYSIS

Project Report on

“Climate Change Impact Analyzer on Crop Yield Patterns”

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR THE DEGREE OF

BACHELOR OF TECHNOLOGY

IN

**Computer Engineering
(Software Engineering)**

OF

VISHWAKARMA INSTITUTE OF TECHNOLOGY

Savitribai Phule Pune University

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UNDER THE GUIDANCE OF

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**DEPARTMENT OF DEPARTMENT OF COMPUTER
ENGINEERING (SOFTWARE ENGINEERING)**

BANSILAL RAMNATH AGARWAL CHARITABLE TRUST'S

VISHWAKARMA INSTITUTE OF TECHNOLOGY

(An Autonomous Institute affiliated to Savitribai Phule Pune University)

2025 - 2026

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CERTIFICATE

This is to certify that the Course Project titled **Climate Change Impact Analyzer on Crop Yield Patterns** submitted by **Pranusha Birhade (12520069)** is in partial fulfillment for the award of Degree of Bachelor of Technology in **Computer Engineering (Software Engineering)** of Vishwakarma Institute of Technology, Savitribai Phule Pune University. This project report is a record of bonafide work carried out by him/her under my guidance during the academic year 2025-26.

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Prof. M.E.Sanap

HOD, CSSE
Prof. Dr.Sunil Sangve

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This is to certify that the Course Project titled **Climate Change Impact Analyzer on Crop Yield Patterns** submitted by **Aditi Avhad (1252080015)** is in partial fulfillment for the award of Degree of Bachelor of Technology in **Computer Engineering (Software Engineering)** of **Vishwakarma Institute of Technology**, Savitribai Phule Pune University. This project report is a record of bonafide work carried out by him/her under my guidance during the academic year 2025-26.

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This is to certify that the Course Project titled **Climate Change Impact Analyzer on Crop Yield Patterns** submitted by **Dharati Biradar (1252080017)** is in partial fulfillment for the award of Degree of Bachelor of Technology in **Computer Engineering (Software Engineering)** of **Vishwakarma Institute of Technology**, Savitribai Phule Pune University. This project report is a record of bonafide work carried out by him/her under my guidance during the academic year 2025-26.

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Academic Year : 2025 - 26
Project Title : **Climate Change Impact Analyzer on Crop Yield Patterns**
Project Area : Sustainability & Green Analytics
Internal Guide : Prof.M.E.Sanap

Signature of Internal Guide

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1. INTRODUCTION

Project Explanation

Climate change has become one of the most critical global challenges, significantly impacting agriculture and food security. Variations in temperature, rainfall patterns, and seasonal cycles directly influence crop growth and productivity. In recent years, unpredictable weather conditions have made traditional farming practices less reliable, creating a need for data-driven solutions. This project, titled “**Climate Change Impact Analyzer on Crop Prediction using Machine Learning**,” focuses on analyzing how environmental factors affect crop production and provides predictive insights to support sustainable agriculture.

The system utilizes historical agricultural data combined with climate-related parameters such as rainfall, season, and geographical location to build a machine learning model capable of predicting crop yield. By applying advanced data preprocessing techniques and algorithms like Random Forest, the project identifies patterns and relationships between climate variables and crop performance. Additionally, the project incorporates data shredding techniques to convert semi-structured data (JSON) into structured tabular format, making it easier to analyze and integrate into predictive models.

Key Features / Points

1. Predicts crop production based on climate inputs
2. Analyzes impact of rainfall and seasonal variations
3. Uses machine learning algorithms for accurate prediction
4. Performs data preprocessing (cleaning, encoding, transformation)
5. Implements **data shredding** (JSON → structured table)
6. Provides user-friendly web interface for input and output
7. Helps in decision-making for farmers and agricultural planning
8. Supports sustainable and climate-aware farming practices

Overall, this project demonstrates the effective use of machine learning and data analytics in addressing real-world agricultural challenges caused by climate change. By predicting crop production and

analyzing environmental impacts, it provides valuable insights that can improve farming efficiency and resource management. The integration of structured and semi-structured data handling further enhances the system's practical applicability, making it a comprehensive solution for sustainable agriculture in a changing climate.

In modern agriculture, data-driven approaches are becoming essential due to the unpredictability introduced by climate change. This project integrates data analytics and machine learning to create a smart crop prediction system. It not only predicts crop yield but also evaluates the influence of climate conditions, making it a dual-purpose analytical tool.

The system also demonstrates the importance of data preprocessing and transformation. Real-world data often comes in different formats, including semi-structured forms like JSON. Through data shredding techniques, this project converts such data into structured tabular format, ensuring compatibility with machine learning models. This enhances the efficiency and scalability of the system.

This project serves as a bridge between environmental science and data analytics by applying machine learning to agricultural data. It highlights how climate factors directly affect crop production and provides a predictive framework to address these challenges. By combining predictive modeling with data engineering techniques, the system contributes to sustainable agriculture and smarter resource utilization.

2. LITERATURE SURVEY

Agriculture has always been highly dependent on environmental conditions, and with the growing impact of climate change, several research studies and applications have been developed to predict crop yield and support farmers. Traditional crop prediction systems were primarily based on statistical methods and historical averages. These methods considered limited factors such as past production and basic rainfall data. However, they lacked accuracy because they could not handle complex relationships between multiple climate variables like temperature, humidity, and seasonal variations.

With the advancement of technology, modern systems have started using machine learning techniques for crop prediction. Many applications use algorithms such as Linear Regression, Decision Trees, and Random Forest to analyze agricultural datasets. These systems take inputs like soil type, rainfall, crop type, and area to predict yield. Some web-based platforms and government tools also provide advisory services to farmers based on weather forecasts. Despite these advancements, most existing systems focus only on prediction and do not deeply analyze the impact of climate change on agricultural productivity.

Existing Systems

1. Traditional Statistical Models

- Based on historical averages
- Use simple mathematical formulas
- Limited number of input parameters

Problem:

- Low accuracy
- Cannot handle complex climate variations

2. Basic Machine Learning Models

- Use algorithms like Linear Regression, Decision Trees
- Predict crop yield using structured data

Problem:

- Ignore dynamic climate changes
- Limited feature analysis
- Less accurate with real-world variability

3. Weather-Based Advisory Systems

- Provide suggestions based on weather forecasts
- Used by agricultural departments

Problem:

- Do not predict actual crop production
- Lack integration with ML models
- Limited personalization

4. Data Handling Limitations

- Most systems use only structured data (Excel, CSV)

Problem:

- Cannot process semi-structured data (JSON)
- Limited scalability and flexibility

Research Gap (Problems in Existing Systems)

- Lack of focus on climate change impact analysis
- Limited use of advanced ML models for higher accuracy
- Inability to handle semi-structured data (JSON)
- Poor integration of multiple environmental factors
- Lack of user-friendly and interactive systems

Proposed System (Your System Improvement)

The proposed system, “Climate Change Impact Analyzer on Crop Prediction using Machine Learning,” overcomes the limitations of existing approaches by integrating advanced machine learning techniques with climate data analysis.

Improvements Over Existing Systems

- Uses Random Forest algorithm for higher prediction accuracy
- Considers multiple climate factors (rainfall, season, area, crop type)
- Focuses on impact of climate change, not just prediction
- Implements data shredding to convert JSON → structured format
- Handles both structured and semi-structured data
- Provides a web-based interface for easy user interaction
- Generates more reliable and practical insights

3. METHODOLOGY

The methodology of the project “Climate Change Impact Analyzer on Crop Prediction using Machine Learning” involves a structured process to analyze agricultural data and predict crop production based on climate factors.

Initially, the dataset is collected from Excel files containing attributes such as crop type, rainfall, season, area, and production. The collected data is then preprocessed by handling missing values, removing inconsistencies, and converting categorical variables into numerical format using encoding techniques. This step ensures that the data is clean and suitable for machine learning.

Next, data shredding is performed to convert semi-structured JSON data into structured tabular format, allowing efficient integration with the dataset. After preprocessing, a machine learning model (Random Forest) is trained using the prepared data. The dataset is divided into training and testing sets to evaluate the model’s performance.

Finally, the trained model is used to predict crop production based on user inputs. The system also analyzes how climate factors such as rainfall and season impact agricultural output. The complete system is implemented using a web interface, where users can input data and obtain prediction results.

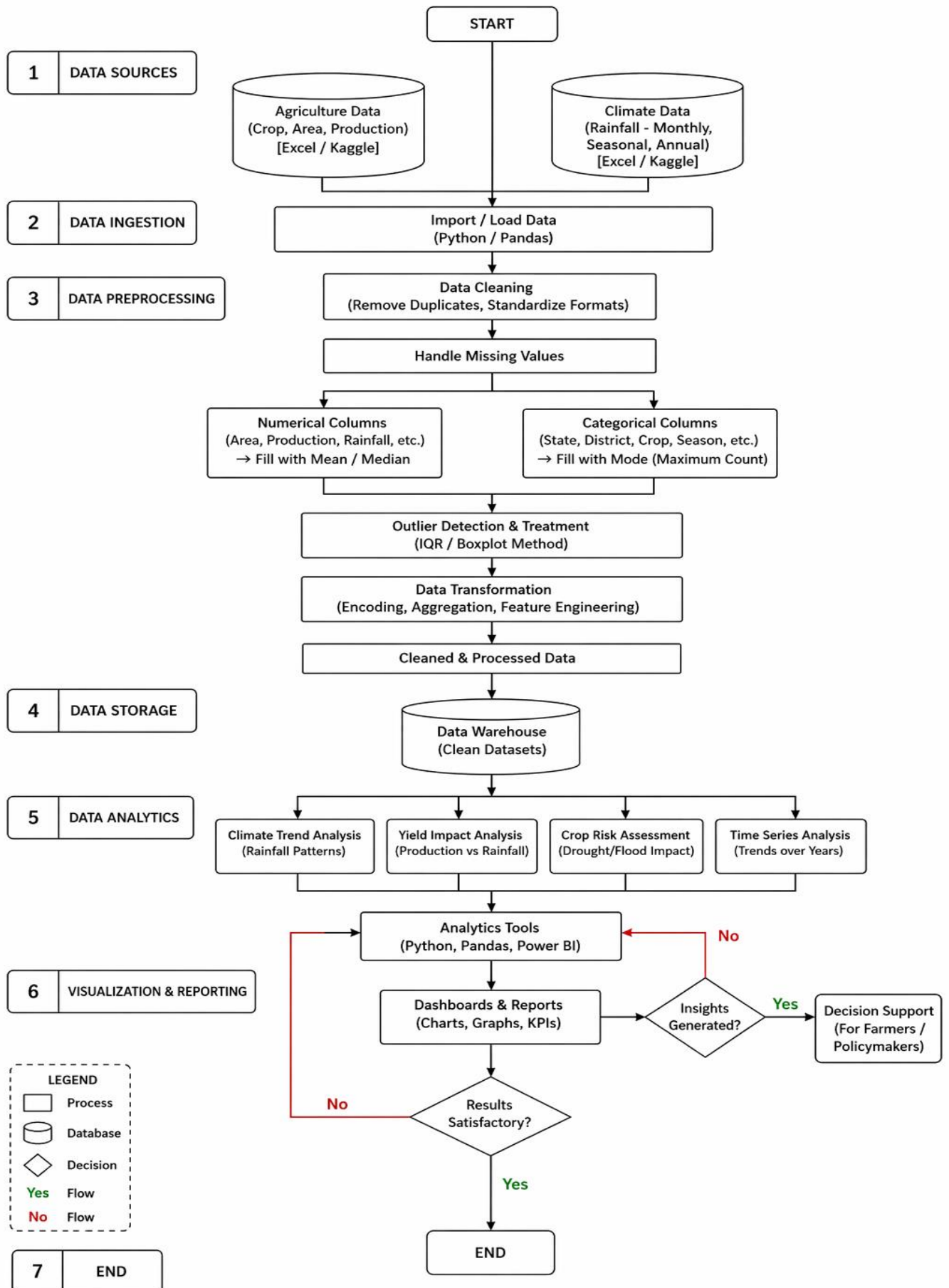
FLOWCHART :

The flowchart shows the complete process of the project, starting from data collection to generating final insights for crop prediction.

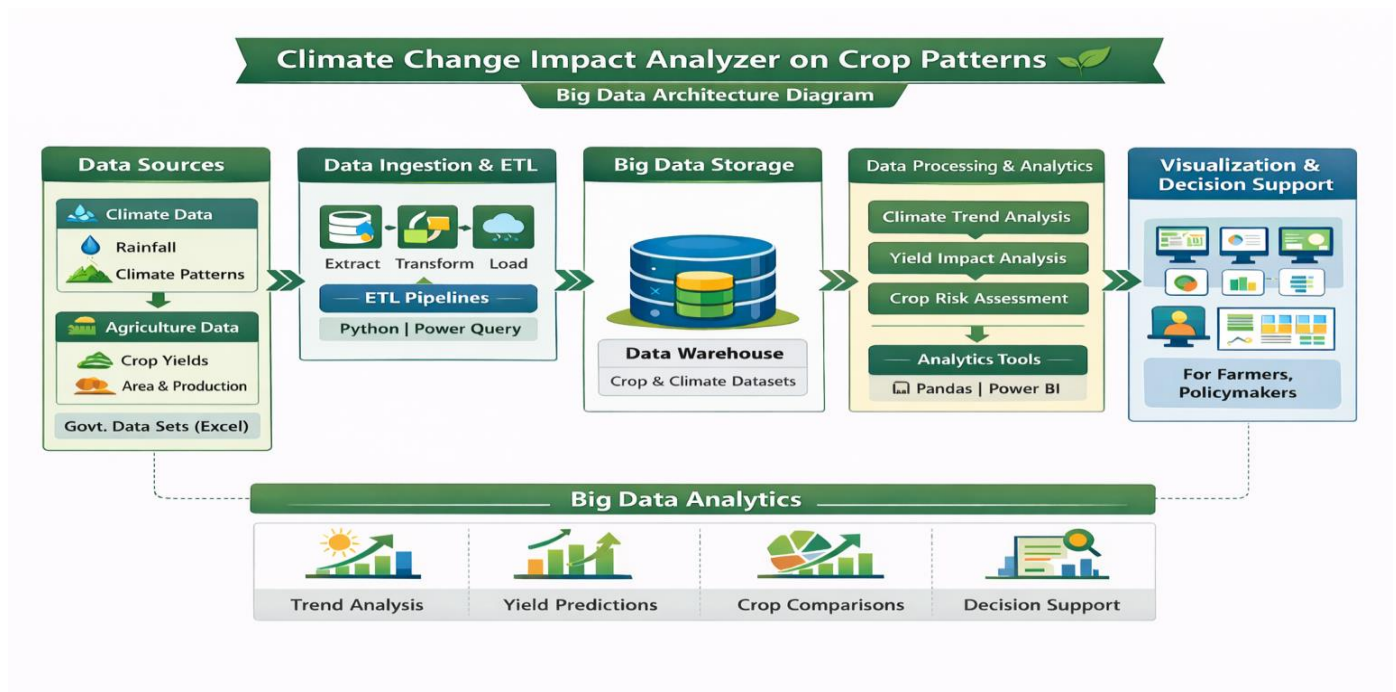
First, data is collected from two sources: agriculture data (crop, area, production) and climate data (rainfall patterns). This data is then imported using Python and Pandas for processing.

Next, in the preprocessing stage, the data is cleaned by removing duplicates and handling missing values. Numerical values are filled using mean/median, while categorical values are filled using mode. Outliers are removed using IQR methods, and data is transformed through encoding and feature engineering to make it suitable for analysis.

The cleaned data is stored in a data warehouse and used for analysis such as climate trends, yield impact, crop risk, and time series analysis. The results are then visualized using charts and dashboards. If useful insights are generated, they help in decision-making for farmers and policymakers; otherwise, the process is repeated.



SYSTEM ARCHITECTURE:



The system architecture represents how data flows through different stages in the Climate Change Impact Analyzer on Crop Prediction system.

1. Data Sources

The system collects data from:

- Climate Data (rainfall, climate patterns)
- Agriculture Data (crop yield, area, production)
- Government datasets (Excel files)

2. Data Ingestion & ETL

Data is processed using ETL (Extract, Transform, Load):

- Extract data from sources
- Transform it into usable format
- Load into system using tools like Python and Power Query

3. Big Data Storage

Processed data is stored in a Data Warehouse, which contains cleaned crop and climate datasets for further analysis.

4. Data Processing & Analytics

Data is analyzed using:

- Climate Trend Analysis
- Yield Impact Analysis
- Crop Risk Assessment

Tools used: Pandas, Power BI

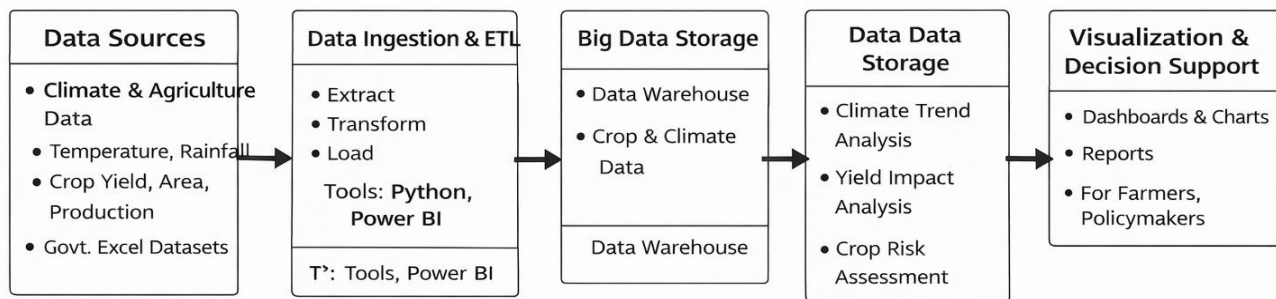
5. Visualization & Decision Support

Results are shown through:

- Dashboards, charts, and reports
- Helps farmers and policymakers in decision-making

SYSTEM WORKFLOW :

Climate Change Impact Analyzer on Crop Patterns



The system collects climate and agriculture data such as rainfall, temperature, crop yield, and area from Excel datasets. This data is processed using ETL (Extract, Transform, Load) techniques with tools like Python.

The processed data is stored in a data warehouse, where it is analyzed to identify climate trends, crop yield impact, and risk assessment. Finally, the results are visualized through dashboards and reports, helping farmers and policymakers make better decisions.

The system collects climate and agricultural data such as rainfall, temperature, crop yield, and area from datasets. This data is processed using ETL techniques and stored in a data warehouse for analysis of trends, yield impact, and crop risk. Finally, the results are visualized through dashboards and reports to support decision-making for farmers and policymakers.

4. FEATURES PROVIDED BY AN APPLICATION

The application “Climate Change Impact Analyzer on Crop Prediction” provides several important features that help users analyze climate effects and predict crop production efficiently.

Steps / Features Explanation

1. Data Input Feature

The application allows users to enter inputs such as crop name, rainfall, season, and area through a user-friendly interface. This ensures easy interaction for users like farmers or researchers.

2. Data Preprocessing

The system automatically cleans the data by handling missing values, removing duplicates, and converting categorical data into numerical form. This improves the quality and accuracy of predictions.

3. Crop Prediction Feature

The core feature of the application is predicting crop production using a machine learning model (Random Forest). Based on user inputs, the system generates an estimated yield.

4. Climate Impact Analysis

The application analyzes how climate factors such as rainfall and seasonal changes affect crop production. This helps in understanding environmental influence on agriculture.

5. Data Shredding (JSON Handling)

The system can process semi-structured data (JSON) and convert it into structured tabular format. This makes it easier to use different types of data in the model.

6. Data Storage

Processed data is stored in a structured format (data warehouse), allowing efficient retrieval and future analysis.

7. Visualization & Reporting

The application presents results using charts, graphs, and dashboards. This makes the output easy to understand and interpret.

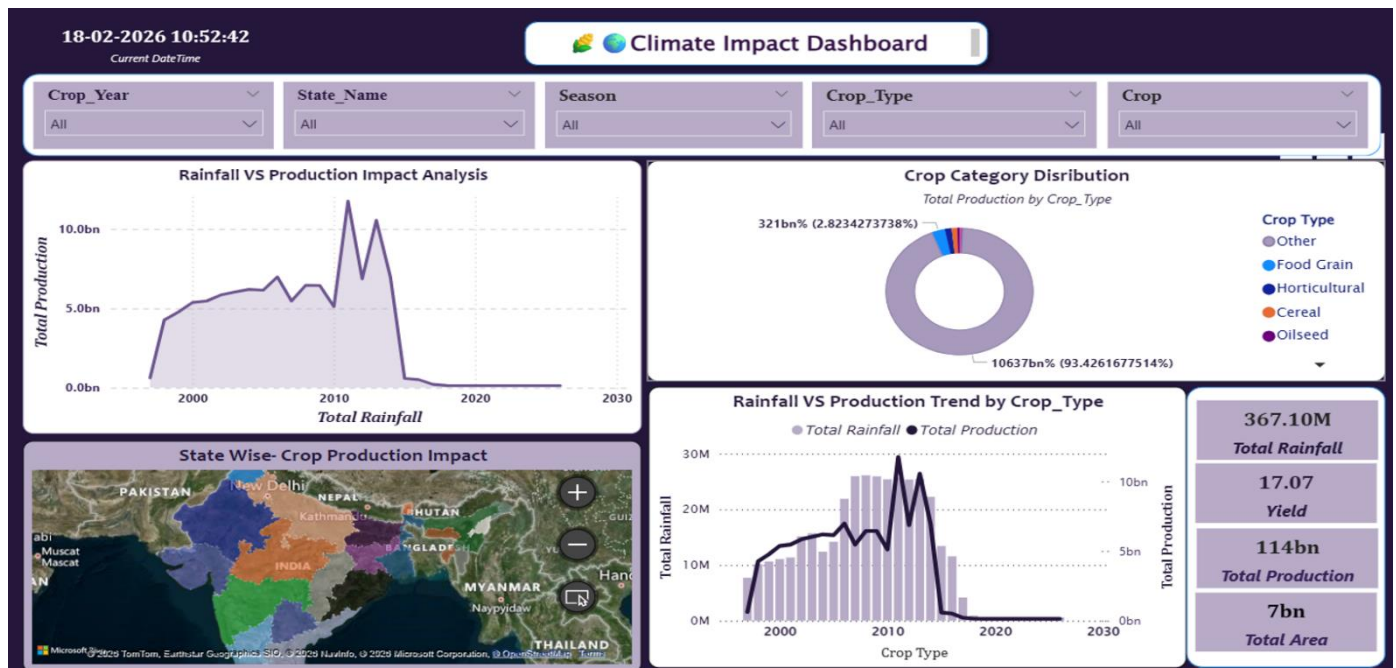
5. USE OF DATASET

1. Climate dataset is used to study temperature and rainfall
2. Crop dataset is used to analyze crop yield
3. Past data helps to find patterns and trends
4. Data is combined to understand climate impact on crops
5. Results are shown using charts and graphs

Dataset Used

1. The dataset contains both climate and crop data
2. Climate data includes temperature and rainfall
3. Crop data includes crop type, season, and yield
4. Data is matched by year, season, and region

6. VISUALS/FLOW OF SYSTEM-POWER BI



Dashboard 1: Climate Trend Analysis

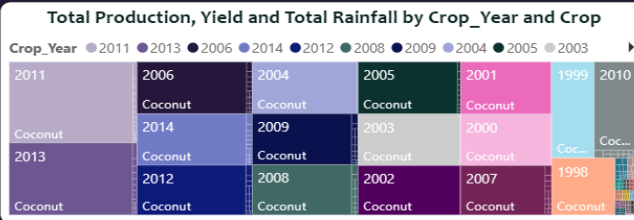
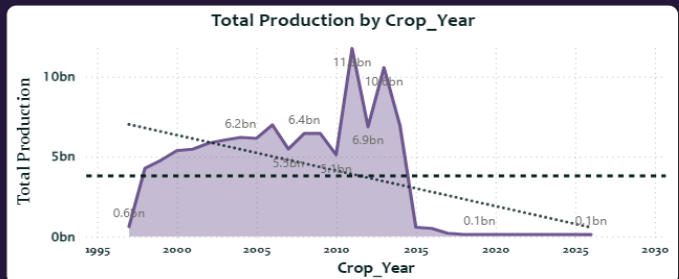
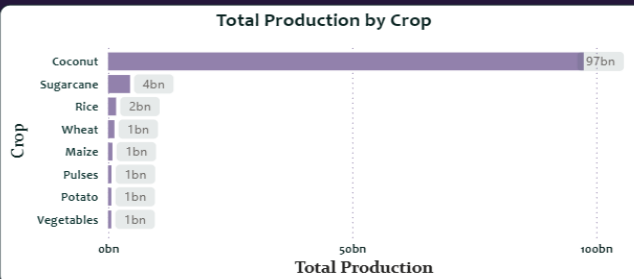
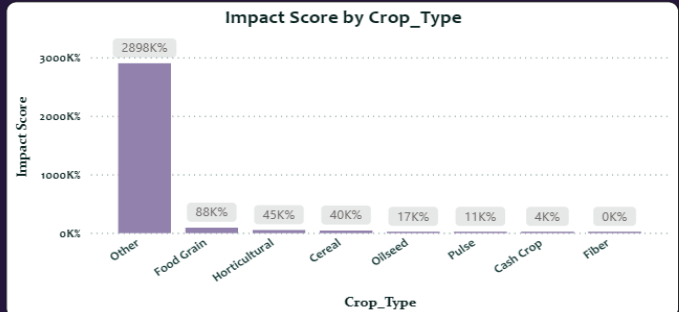
Shows rainfall patterns over different seasons and years.

Helps understand how climate changes over time.

18-02-2026 10:52:42

Current DateTime

Climate Impact Deep Analysis

**3101.57K%**
Impact Score**310.157**
Climate Impact Efficiency

Dashboard 2: Crop Yield Analysis

Displays crop production based on factors like area, season, and rainfall.

Helps compare yield variations across crops and regions.

18-02-2026 10:52:42

Current DateTime

Histogram of Crop Production

Histogram of Crop Production

450K

Avg Production

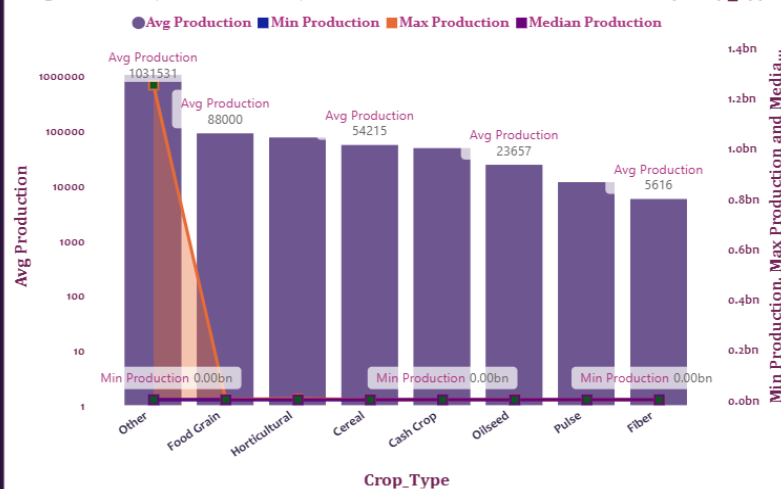
8.74K

Median Production

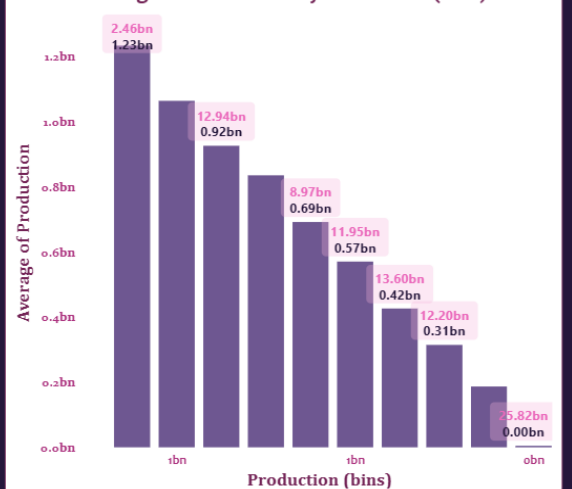
1bn

Max Production

Avg Production, Min Production, Max Production and Median Production by Crop_Type



Average of Production by Production (bins)



Dashboard 3: Impact & Risk Analysis

Shows how climate factors affect crop production and identifies high/low yield conditions.

Helps in decision-making and risk assessment.

7. PREDICTION MODEL

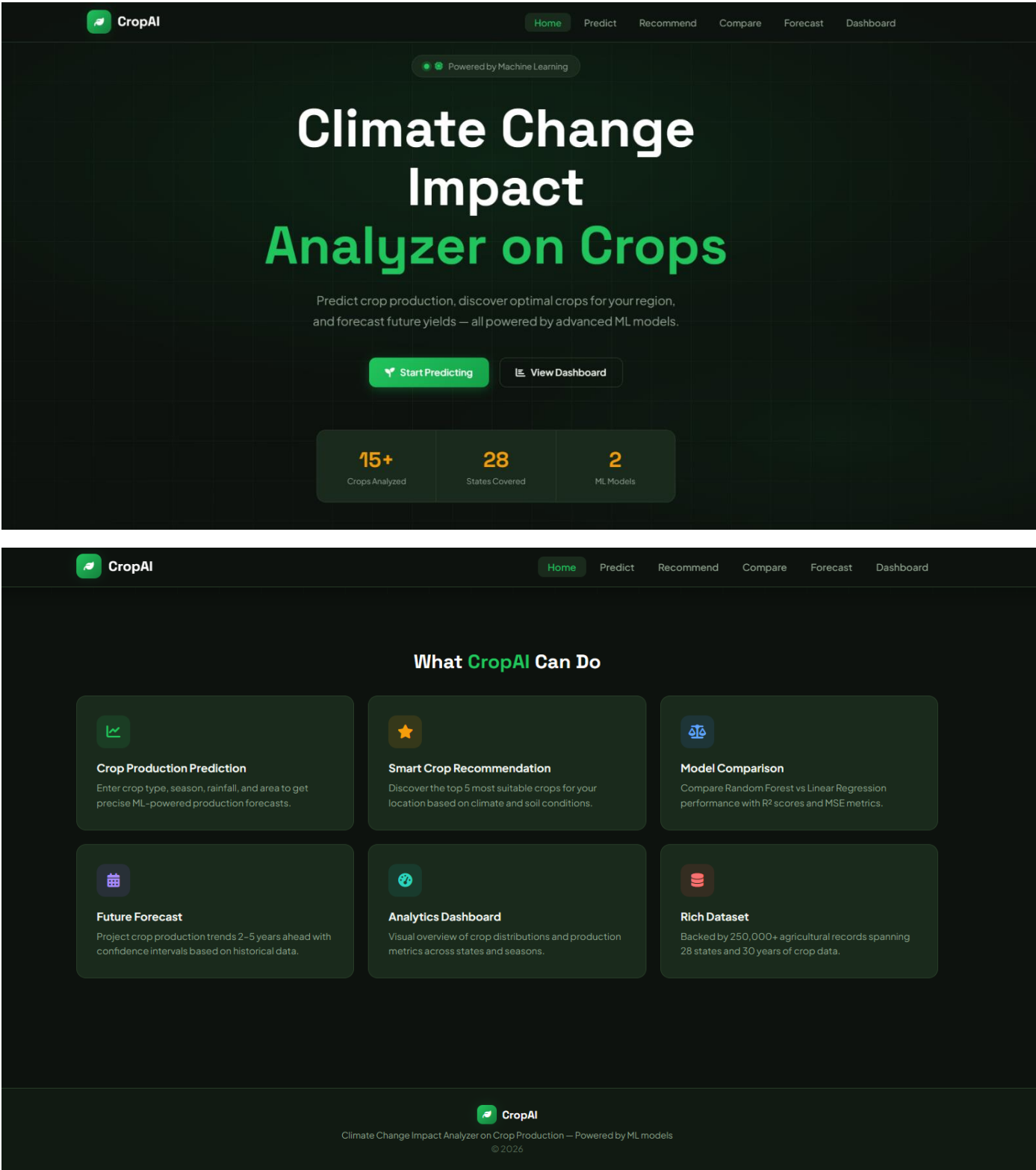
- The system uses a machine learning model (Random Forest) for prediction
- Input features: crop, rainfall, season, area
- Model is trained on historical agricultural data
- Data is split into training and testing for accuracy
- The model learns patterns between climate factors and crop production
- Output: predicted crop yield

8. COMPARATIVE ANALYSIS

- Compares actual vs predicted crop production
- Evaluates model performance using accuracy/error metrics
- Analyzes how different climate factors affect yield
- Helps identify which conditions give high or low production
- Useful for improving model performance and decision-making

9. SCREENSHOTS OF RESULTS

HOME PAGE

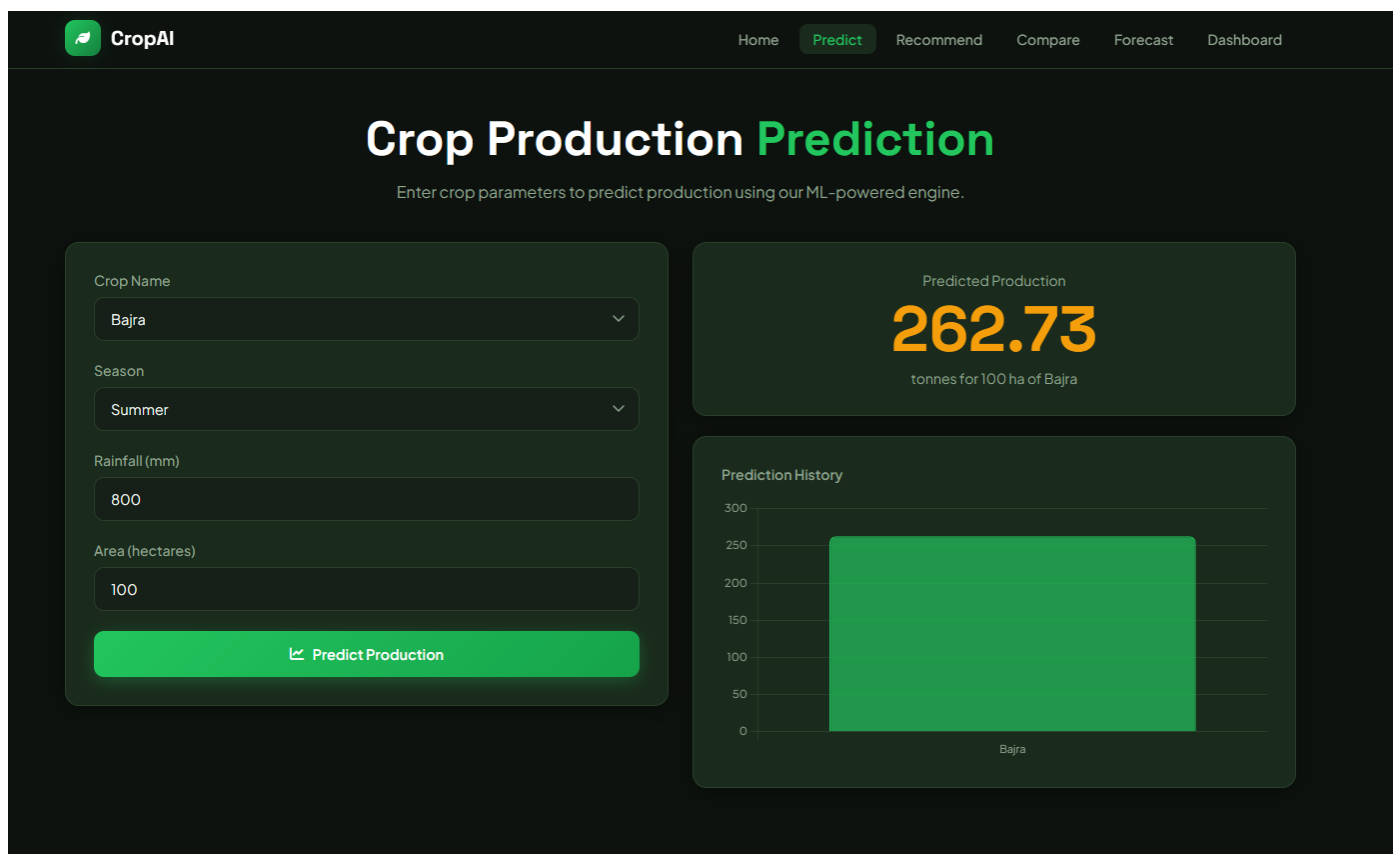


Explanation

The home page of the CropAI system provides an overview of the application, highlighting its purpose of analyzing climate change impact on crop production using machine learning. It includes navigation options like Predict, Recommend, Compare, Forecast, and Dashboard for easy access to different features.

It displays key functionalities such as crop prediction, smart crop recommendation, model comparison, future forecasting, analytics dashboard, and dataset information.

PREDICTION PAGE :



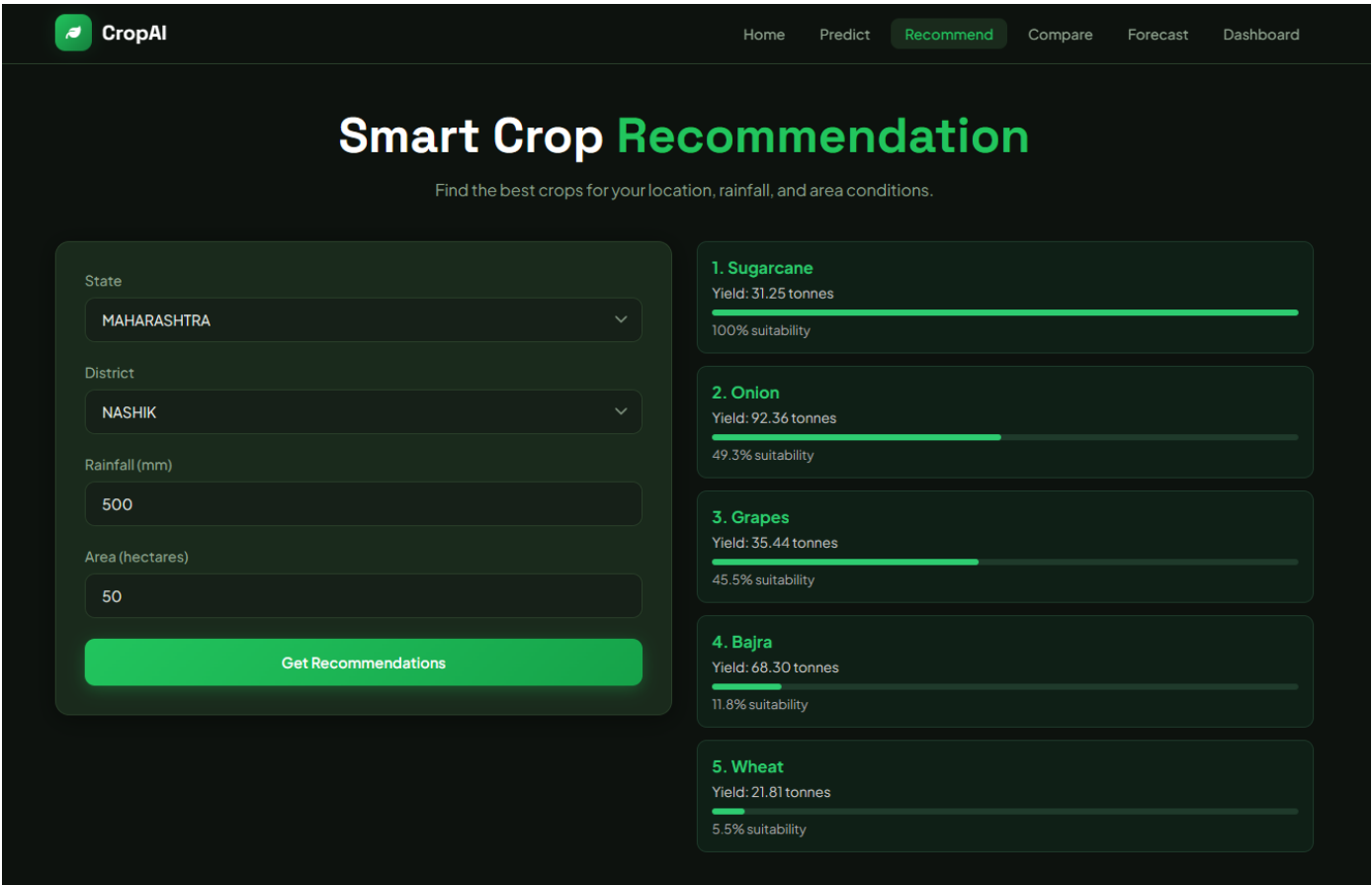
The screenshot shows the 'Crop Production Prediction' page of the CropAI system. The page has a dark theme with green accents. At the top, there is a navigation bar with links: Home, Predict (active), Recommend, Compare, Forecast, and Dashboard. The main heading is 'Crop Production Prediction' in white and green, with a subtitle 'Enter crop parameters to predict production using our ML-powered engine.' Below this, there are four input fields: 'Crop Name' (Bajra), 'Season' (Summer), 'Rainfall (mm)' (800), and 'Area (hectares)' (100). A green button labeled 'Predict Production' is at the bottom of the input section. To the right, a large green box displays the 'Predicted Production' as '262.73 tonnes for 100 ha of Bajra'. Below this, a 'Prediction History' section shows a bar chart with a single green bar for 'Bajra' reaching a value of approximately 260 on a scale from 0 to 300.

Prediction Page Explanation

The prediction page allows users to input crop details such as crop name, season, rainfall, and area to predict crop production using a machine learning model. After entering the values and clicking “Predict Production,” the system generates the estimated crop yield.

The page also displays the predicted output clearly along with a simple chart showing prediction history. This helps users easily understand the results and analyze crop production based on climate conditions.

RECOMMENDATION PAGE :



Recommendation Page Explanation

The recommendation page suggests the most suitable crops based on user inputs such as state, district, rainfall, and area. After entering the details, the system analyzes climate conditions and provides a list of recommended crops.

Each crop is displayed with its predicted yield and suitability percentage, helping users identify the best options for cultivation. This feature supports better crop selection and improves agricultural planning.

MODEL COMPARISON :

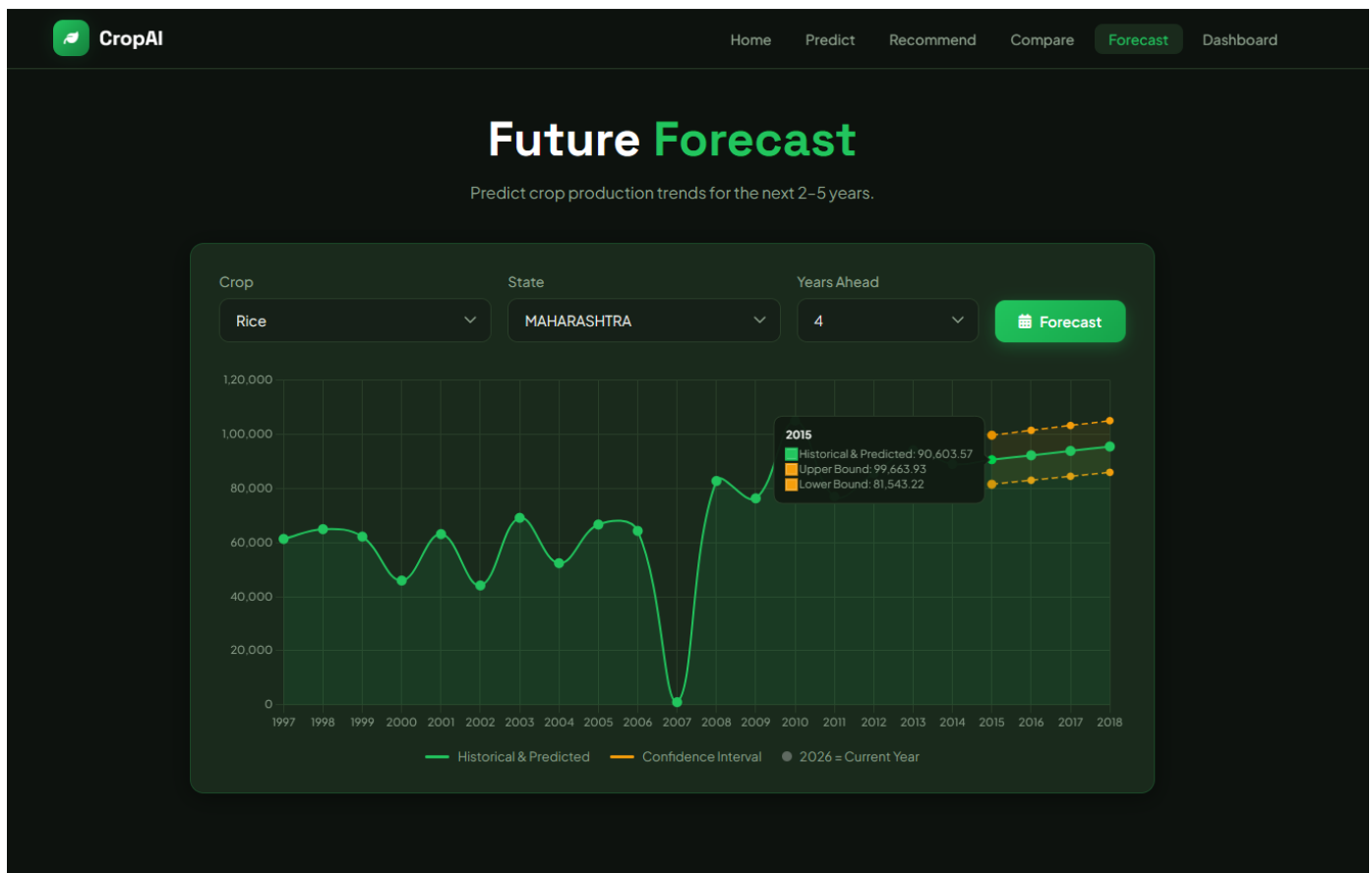


Model Comparison Page Explanation :

The model comparison page compares the performance of two machine learning models: Random Forest and Linear Regression. Users enter inputs like crop, rainfall, and area, and the system evaluates both models.

The comparison is shown using metrics such as Mean Squared Error (MSE) and R^2 Score, where lower MSE and higher R^2 indicate better performance. This helps identify which model gives more accurate crop predictions.

FUTURE FORECAST PAGE :

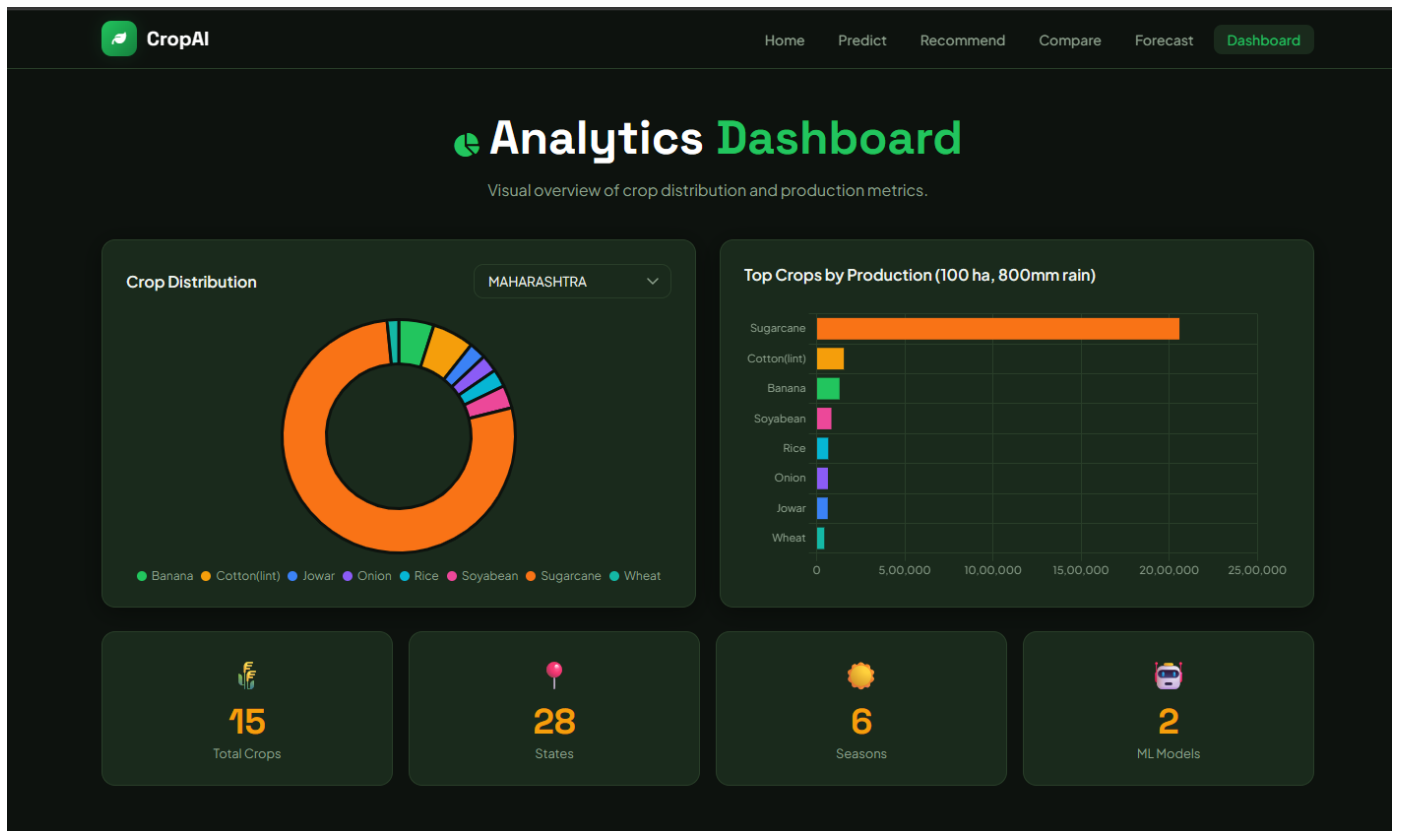


Future Forecast Page Explanation

The future forecast page predicts crop production trends for the next few years based on historical data. Users select crop, state, and number of years, and the system generates future predictions.

The results are displayed using a line graph showing past data and future estimates along with confidence intervals. This helps users understand future trends and plan agricultural activities effectively.

DASHBOARD :



Analytics Dashboard Explanation

The analytics dashboard provides a visual overview of crop production and distribution data. It displays charts such as crop distribution and top crops by production, helping users understand patterns across different crops and regions.

It also shows key statistics like total crops, number of states, seasons, and ML models used. This dashboard helps in quick analysis and supports better decision-making.

10.ADVANTAGES AND LIMITATIONS

Advantages

- Provides accurate crop prediction using machine learning
- Helps understand the impact of climate change on agriculture
- Supports better decision-making for farmers and policymakers
- Handles both structured (Excel) and semi-structured (JSON) data
- Includes data visualization dashboards for easy understanding
- User-friendly web interface for easy interaction
- Saves time and reduces manual analysis

Limitations

- Prediction accuracy depends on quality and size of dataset
- Limited to available features (rainfall, season, etc.)
- Does not consider all factors like soil type, pests, fertilizers
- Requires internet and technical setup for use
- Model may not perform well for unseen or extreme conditions
- Forecasting is based on past data, so future uncertainty exists

11.FUTURE SCOPE

- Include more parameters like **soil type, temperature, humidity, fertilizers** for better accuracy
- Integrate **real-time weather data APIs** for live predictions
- Improve model using advanced techniques like **Deep Learning**
- Develop a **mobile application** for easy access by farmers
- Expand dataset to cover more **regions and crops**
- Add **disease prediction and pest analysis** features

- Implement **IoT sensors** for real-time field data collection
- Provide **multi-language support** for wider usability

CONCLUSION

The project “**Climate Change Impact Analyzer on Crop Prediction using Machine Learning**” demonstrates how machine learning can be effectively used to analyze climate factors and predict crop production. By using data such as rainfall, season, and area, the system provides accurate predictions and helps understand the impact of climate change on agriculture.

The project includes important processes like data preprocessing, data shredding, model building, and visualization, making it a complete analytical system. Features such as crop prediction, recommendation, forecasting, and dashboards enhance its practical usability.

Although the system depends on data quality and limited parameters,

It provides a strong foundation for improving agricultural decision-making. Overall, the project offers a useful and scalable solution for supporting sustainable farming and better resource management.

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